

GCSE Mathematics - Foundation Tier

Paper 3 (Calculator) Practice - Set 10

Time: 1 hour 30 minutes | Total marks: 80 | Calculator allowed

Instructions

- Use black ink or ball-point pen.
- Answer ALL questions in the spaces provided.
- You must show all your working - answers given without working may not score full marks.
- Diagrams are NOT accurately drawn unless otherwise stated.
- If your calculator does not have a pi button, use $\pi = 3.142$.

1.

Write 0.08 as a percentage.

(Total for Question 1 is 1 marks)

2.

Work out 60% of £450.

(Total for Question 2 is 2 marks)

3.

- (a) Find the value of 2^5 . (1)
- (b) Find the value of $\sqrt{225}$. (1)

(Total for Question 3 is 2 marks)

4.

Write the ratio 36 : 48 : 60 in its simplest form.

(Total for Question 4 is 2 marks)

5.

The bar chart shows the type of pet kept by 80 students.

[Diagram: bar chart of pets owned by 80 students. Bar heights - Dog: 26, Cat: 22, Rabbit: 14, Fish: 10, Hamster: 8.]

- (a) How many more students keep a cat than a hamster? (1)
- (b) Work out the fraction of students who keep a dog. Give your answer in its simplest form. (2)

(Total for Question 5 is 3 marks)

6.

Solve $4x + 3 = 19$.

(Total for Question 6 is 2 marks)

7.

A right-angled triangle has shorter sides 3.6 cm and 4.8 cm. Work out the length of the hypotenuse.

[Diagram: right-angled triangle with shorter sides 3.6 cm and 4.8 cm; hypotenuse labelled.]

(Total for Question 7 is 3 marks)

8.

The frequency table shows the points scored by a basketball team in 20 games.

Points	50-59	60-69	70-79	80-89	90-99
Freq.	2	6	7	4	1

(a) Write down the modal class. (1)

(b) Using class midpoints, estimate the mean points scored per game. Give your answer to 1 decimal place. (2)

(Total for Question 8 is 3 marks)

9.

A walker leaves point P and walks 6 km on a bearing of 060 deg to point Q. From Q, she walks 8 km on a bearing of 150 deg to point R.

[Diagram: point P with north line; PQ 6 km on bearing 060 deg; point Q with its north line; QR 8 km on bearing 150 deg.]

(a) Work out the angle PQR. (2)

(b) Hence describe triangle PQR (or angle PQR). (1)

(Total for Question 9 is 3 marks)

10.

The formula $A = \pi \times r^2$ gives the area of a circle.

(a) Work out A when $r = 4$ cm. Use $\pi = 3.142$. Give your answer to 1 decimal place. (2)

The formula $C = 2 \times \pi \times r$ gives the circumference of a circle.

(b) Work out C when $r = 7$ cm. Use $\pi = 3.142$. Give your answer to 1 decimal place. (1)

(Total for Question 10 is 3 marks)

11.

A cube has a total surface area of 96 cm^2 .

(a) Work out the length of one side of the cube. (2)

(b) Hence work out the volume of the cube. (1)

12.

Hugo invests £2200 at 3% compound interest per year.

Work out the value after 3 years. Give your answer to the nearest penny.

(Total for Question 12 is 4 marks)

13.

A bag contains 4 red counters and 6 blue counters. A counter is taken at random and replaced. A second counter is then taken at random.

- (a) Write down the probability of a single counter being red. (1)
- (b) Work out the probability of getting one red counter and one blue counter, in either order. (3)

(Total for Question 13 is 4 marks)

14.

A rectangular field has a width of 35 m. A path runs diagonally across the field, from one corner to the opposite corner, and has length 53 m.

[Diagram: rectangle of width 35 m, length L , and a diagonal of 53 m drawn corner-to-corner.]

- (a) Show that $L^2 = 1584$. (3)
- (b) Hence work out the length of the field. Give your answer to 1 decimal place. (2)

(Total for Question 14 is 5 marks)

15.

A solid cylinder has a radius of 6 cm and a height of 14 cm.

[Diagram: solid cylinder of radius 6 cm and height 14 cm.]

- (a) Work out the volume. Use $\pi = 3.142$. Give your answer to 1 decimal place. (3)
- (b) Work out the total surface area. Give your answer to 1 decimal place. (2)

(Total for Question 15 is 5 marks)

16.

The n th term of a sequence is given by $T_n = 4n - 3$.

- (a) Find the 1st and 2nd terms. (2)
- (b) Show that the sum of the 1st 10 terms is 190. (2)

(Total for Question 16 is 4 marks)

17.

The table shows the average daily rainfall (mm) and the daily ice-cream sales (£) at a seaside kiosk over 8 days.

Rain (mm)	0	2	4	6	8	10	12	15
Sales (£)	320	290	260	220	180	150	120	80

- On the grid provided, plot these data as a scatter graph. (2)
- Describe the correlation. (1)
- Draw a line of best fit. (1)
- Estimate the daily ice-cream sales when daily rainfall is 7 mm. (1)

(Total for Question 17 is 5 marks)

18.

A swimming pool is in the shape of a cuboid 25 m long, 12 m wide and 2 m deep.

- Work out the volume of the pool. Give your answer in m^3 . (2)
- Given that $1 \text{ m}^3 = 1000$ litres, convert the volume to litres. (1)
- A pump fills the pool at a rate of 500 litres per minute. Work out how long it takes to fill the pool. Give your answer in hours. (2)

(Total for Question 18 is 5 marks)

19.

A scale drawing of a garden uses a scale of 1 : 200.

- Two flower beds are 6 cm apart on the drawing. Work out the real distance between them in metres. (2)
- A path is 15 m long in real life. Work out its length on the drawing in centimetres. (2)
- A circular pond has a diameter of 4 m in real life. Work out its diameter on the drawing in centimetres. (1)

(Total for Question 19 is 5 marks)

20.

The expression for the perimeter, P , of a regular pentagon of side length s is $P = 5s$.

The pentagon has perimeter 60 cm.

- Work out the side length s . (2)

A regular hexagon (six equal sides) has side length $0.8s$, where s is the side length of the pentagon above.

- Show that the perimeter of the hexagon is 57.6 cm. (3)

(Total for Question 20 is 5 marks)

21.

Two savings plans are compared.

Plan P: £200 deposited every year for 3 years, with no interest paid until the end.

Plan Q: £600 deposited in one lump sum at the start of year 1, earning 4% compound interest per year.

- (a) Work out the total in Plan P at the end of 3 years. (2)
- (b) Work out the total in Plan Q at the end of 3 years. Give your answer to the nearest penny. (3)
- (c) Which plan is worth more after 3 years? By how much? (1)

(Total for Question 21 is 6 marks)

22.

70 people at a sports centre were asked which activity they took part in. The two-way table shows some information.

	Gym	Swim	Yoga	Total
Adults	—	16	—	50
Children	8	—	—	20
Total	28	22	—	70

- (a) Complete the two-way table. (3)
- (b) A person is chosen at random. Work out the probability that the person is an adult who took part in Yoga. (2)

(Total for Question 22 is 5 marks)

TOTAL FOR PAPER: 80 MARKS

Mark Scheme

Foundation Tier - Paper 3 (Calculator) - Set 10 | 80 marks

Question 1

- 8 (%) - B1

Total: 1 marks

Question 2

- 450×0.6 - M1
- £270 - A1

Total: 2 marks

Question 3

- (a) 32 - B1
- (b) 15 - B1

Total: 2 marks

Question 4

- All three divided by a common factor (e.g. /12 $\rightarrow$ 3:4:5) - M1
- 3 : 4 : 5 - A1

Total: 2 marks

Question 5

- (a) $22 - 8 = 14$ - B1
- (b) 26 / 80 simplified - M1
- (b) $13/40$ - A1

Total: 3 marks

Question 6

- $4x = 16$ - M1
- $x = 4$ - A1

Total: 2 marks

Question 7

- $h^2 = 3.6^2 + 4.8^2 (= 12.96 + 23.04)$ - M1
- $= 36$ - M1
- $h = 6$ (cm) - A1

Total: 3 marks

Question 8

- (a) 70 - 79 - B1
- (b) $\text{Sum}(f \times \text{midpoint}) = 54.5 \times 2 + 64.5 \times 6 + 74.5 \times 7 + 84.5 \times 4 + 94.5 \times 1 = 109 + 387 + 521.5 + 338 + 94.5 = 1450; / 20 = 72.5$ - M1
- (b) 72.5 - A1

Total: 3 marks

Question 9

- (a) Back-bearing at Q for PQ is $60 + 180 = 240$ deg; QR bearing is 150 deg; angle = $240 - 150$ - M1
- (a) 90 (deg) - A1
- (b) Right-angled (at Q) - B1

Total: 3 marks

Question 10

- (a) 3.142×16 attempted - M1
- (a) 50.3 (cm²) - A1
- (b) $2 \times 3.142 \times 7 = 43.988 \rightarrow 44.0$ (cm) - B1

Total: 3 marks

Question 11

- (a) $96 / 6 = 16$ (area of one face) - M1
- (a) $\sqrt{16} = 4$ (cm) - A1
- (b) $4^3 = 64$ (cm³) - B1

Total: 3 marks

Question 12

- Multiplier 1.03 seen - M1
- $2200 \times 1.03^2 (= 2333.98)$ seen - M1
- 2200×1.03^3 - M1
- £2404.00 (accept £2404.0 - £2404.01) - A1

Total: 4 marks

Question 13

- (a) $4/10$ (or $2/5$) - B1
- (b) RB and BR both considered - M1
- (b) $(4/10)(6/10) + (6/10)(4/10)$ or $2 \times 4/10 \times 6/10$ - M1
- (b) $48/100 = 12/25$ (accept 0.48) - A1

Total: 4 marks

Question 14

- (a) $L^2 + 35^2 = 53^2$ (Pythagoras applied) - M1
- (a) 2809 - 1225 evaluated - M1
- (a) $L^2 = 1584$ shown - C1
- (b) $L = \sqrt{1584}$ attempted - M1
- (b) 39.8 (m) - A1

Total: 5 marks

Question 15

- (a) $\pi \times 6^2 \times 14 (= 3.142 \times 36 \times 14)$ - M1
- (a) 1583.568 - M1

- (a) $1583.6 \text{ (cm}^3\text{)}$ accept $1583.4 - 1583.8$ - A1
- (b) $2 \times \pi \times 6^2 + 2 \times \pi \times 6 \times 14 = 226.224 + 527.856$ - M1
- (b) $754.1 \text{ (cm}^2\text{)}$ accept $753.9 - 754.3$ - A1

Total: 5 marks

Question 16

- (a) $T_1 = 1; T_2 = 5$ - B2 (B1 for either correct)
- (b) $\text{Sum} = T_1 + T_2 + \dots + T_{10} = 1 + 5 + 9 + 13 + 17 + 21 + 25 + 29 + 33 + 37 = 190$ - M1
- (b) 190 shown - A1

Total: 4 marks

Question 17

- (a) All 8 points plotted correctly - B2 (B1 for 6+)
- (b) Negative correlation (strong) - B1
- (c) Sensible line of best fit - B1
- (d) Sales at 7 mm rain (accept $\pounds 190 - \pounds 220$) - B1

Total: 5 marks

Question 18

- (a) $25 \times 12 \times 2$ - M1
- (a) $600 \text{ (m}^3\text{)}$ - A1
- (b) $600 \times 1000 = 600\,000$ litres - B1
- (c) $600\,000 / 500 = 1200$ minutes; $/ 60 = 20$ hours - M1
- (c) 20 hours - A1

Total: 5 marks

Question 19

- (a) $6 \times 200 = 1200 \text{ cm}$ - M1
- (a) 12 (m) - A1
- (b) $15 \text{ m} = 1500 \text{ cm}; / 200 = 7.5 \text{ cm}$ - M1
- (b) 7.5 (cm) - A1
- (c) $4 \text{ m} = 400 \text{ cm}; / 200 = 2 \text{ cm}$ - B1

Total: 5 marks

Question 20

- (a) $60 / 5$ - M1
- (a) $s = 12 \text{ (cm)}$ - A1
- (b) Hex side $= 0.8 \times 12 = 9.6 \text{ (cm)}$ - M1
- (b) Perimeter of hex $= 6 \times 9.6$ - M1
- (b) 57.6 (cm) shown - A1

Total: 5 marks

Question 21

- (a) 200×3 - M1
- (a) $\pounds 600$ - A1
- (b) Multiplier 1.04 seen - M1

- (b) $600 \times 1.04^2 (= 648.96)$ seen - M1
- (b) $600 \times 1.04^3 = £674.92$ (accept $£674.916 - £674.92$) - A1
- (c) Q is worth more by $674.92 - 600 = £74.92$ (accept $£74.92$) - B1

Total: 6 marks

Question 22

- (a) Children Swim = $22 - 16 = 6$; Children Yoga = $20 - 8 - 6 = 6$ - M1
- (a) Adult Gym = $28 - 8 = 20$; Adult Yoga = $50 - 20 - 16 = 14$ - M1
- (a) Yoga column total = $14 + 6 = 20$; grand total 70 confirmed - A1
- (b) Adult Yoga = 14 out of 70 - M1
- (b) $14/70 = 1/5$ - A1

Total: 5 marks

Worked Solutions

Foundation Tier - Paper 3 (Calculator) - Set 10 | Detailed explanations

Question 1

To convert a decimal to a percentage, multiply by 100.

$$0.08 \times 100 = 8\%.$$

Question 2

$$60\% \text{ of } 450 = 0.6 \times 450 = 270.$$

Answer: £270.

Question 3

$$(a) 2^5 = 2 \times 2 \times 2 \times 2 \times 2 = 32.$$

$$(b) \sqrt{225} = 15 \text{ (since } 15 \times 15 = 225\text{)}.$$

Question 4

HCF of 36, 48, 60 is 12.

$$\text{Divide each: } 36/12 = 3, 48/12 = 4, 60/12 = 5.$$

Answer: 3 : 4 : 5.

Question 5

$$(a) \text{Cat } 22 - \text{Hamster } 8 = 14.$$

$$(b) \text{Dog } 26 \text{ out of } 80. \text{ HCF of } 26 \text{ and } 80 \text{ is } 2, \text{ so } 26/80 = 13/40.$$

Question 6

$$4x + 3 = 19. \text{ Subtract } 3: 4x = 16. \text{ Divide by } 4: x = 4.$$

Question 7

$$h^2 = 3.6^2 + 4.8^2 = 12.96 + 23.04 = 36.$$

$$h = \sqrt{36} = 6 \text{ cm.}$$

Question 8

(a) Highest frequency is 7, in 70-79. Modal class: 70-79.

(b) Use class midpoints: 54.5, 64.5, 74.5, 84.5, 94.5.

$$\text{Sum}(fx) = 54.5 \times 2 + 64.5 \times 6 + 74.5 \times 7 + 84.5 \times 4 + 94.5 \times 1 = 109 + 387 + 521.5 + 338 + 94.5 = 1450.$$

$$\text{Mean} = 1450 / 20 = 72.5 \text{ (1 dp).}$$

Question 9

(a) At Q, the back-bearing of PQ is $60 + 180 = 240$ deg.

The bearing of QR is 150 deg.

Angle PQR = $240 - 150 = 90$ deg.

(b) Triangle PQR has a right angle at Q.

Question 10

(a) $A = \pi \times r^2 = 3.142 \times 16 = 50.272$, i.e. 50.3 cm^2 to 1 dp.

(b) $C = 2 \times \pi \times r = 2 \times 3.142 \times 7 = 43.988$, i.e. 44.0 cm to 1 dp.

Question 11

(a) A cube has 6 identical square faces. Each face area = $96 / 6 = 16 \text{ cm}^2$.

Side = $\sqrt{16} = 4 \text{ cm}$.

(b) Volume = $\text{side}^3 = 4^3 = 64 \text{ cm}^3$.

Question 12

Multiplier 1.03.

Year 1: $2200 \times 1.03 = \text{£}2266.00$.

Year 2: $2266 \times 1.03 = \text{£}2333.98$.

Year 3: $2333.98 \times 1.03 = \text{£}2403.9994$, i.e. $\text{£}2404.00$ to the nearest penny.

Question 13

(a) $P(\text{red}) = 4/10 = 2/5$.

(b) Two independent picks. "One red and one blue" can be RB or BR.

$P(\text{RB}) = (4/10) \times (6/10) = 24/100$.

$P(\text{BR}) = (6/10) \times (4/10) = 24/100$.

Total = $24/100 + 24/100 = 48/100 = 12/25 (= 0.48)$.

Question 14

(a) The diagonal is the hypotenuse of a right-angled triangle.

By Pythagoras: $L^2 + 35^2 = 53^2$.

$L^2 = 53^2 - 35^2 = 2809 - 1225 = 1584$. (Shown.)

(b) $L = \sqrt{1584} = 39.799\dots$, i.e. 39.8 m to 1 dp.

Question 15

(a) $V = \pi \times r^2 \times h = 3.142 \times 36 \times 14 = 1583.568$, i.e. 1583.6 cm^3 .

(b) Surface area = 2 (circular ends) + curved.

$2 \times \pi \times 6^2 = 2 \times 3.142 \times 36 = 226.224$.

$2 \times \pi \times 6 \times 14 = 527.856$.

Total = 754.08 , i.e. 754.1 cm^2 to 1 dp.

Question 16

- (a) $T_1 = 4(1) - 3 = 1$. $T_2 = 4(2) - 3 = 5$.
(b) Terms T_1 to T_{10} : 1, 5, 9, 13, 17, 21, 25, 29, 33, 37.
Sum = $1+5+9+13+17+21+25+29+33+37 = 190$. (Shown.)
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Question 17

- (a) Plot 8 points.
(b) Negative correlation - more rain, fewer ice-creams sold.
(c) Best fit drawn.
(d) At 7 mm rain, line gives roughly £190 - £220.
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Question 18

- (a) $V = 25 \times 12 \times 2 = 600 \text{ m}^3$.
(b) $600 \times 1000 = 600\,000$ litres.
(c) Time = $600\,000 / 500 = 1200$ minutes = 20 hours.
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Question 19

- (a) $6 \text{ cm} \times 200 = 1200 \text{ cm} = 12 \text{ m}$.
(b) $15 \text{ m} = 1500 \text{ cm}$. On drawing: $1500 / 200 = 7.5 \text{ cm}$.
(c) $4 \text{ m} = 400 \text{ cm}$. On drawing: $400 / 200 = 2 \text{ cm}$.
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Question 20

- (a) $5s = 60$, so $s = 12 \text{ cm}$.
(b) Hexagon side = $0.8 \times 12 = 9.6 \text{ cm}$.
Hexagon perimeter = $6 \times 9.6 = 57.6 \text{ cm}$. (Shown.)
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Question 21

- (a) Plan P: 3 deposits of £200, no interest. Total = $3 \times 200 = £600$.
(b) Plan Q: $600 \times 1.04^3 = 600 \times 1.124864 = 674.9184$, i.e. £674.92 to the nearest penny.
(c) Q is worth more by $674.92 - 600 = £74.92$.
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Question 22

- (a) Children Swim = $22 - 16 = 6$. Children Yoga = $20 - 8 - 6 = 6$.
Adult Gym = $28 - 8 = 20$. Adult Yoga = $50 - 20 - 16 = 14$.
Yoga total = $14 + 6 = 20$. Grand total check: $28 + 22 + 20 = 70$.
(b) Adult AND Yoga = 14 out of 70.
 $P = 14/70 = 1/5$.
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